Lab02_Pierre

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QUESTÃO 1:

As estatísticas que são suficientes são os número de voos com ARRIVAL_DELAY > 10 e o número de voos válidos, agrupados por mês, dia e companhia aérea.

QUESTÃO 2:

```r
library(dplyr)
```

```
Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

    filter, lag

Os seguintes objetos são mascarados por 'package:base':

    intersect, setdiff, setequal, union
```

```r
getStats <- function(input, pos) {
  input %>%
    filter(AIRLINE %in% c("AA", "DL", "UA", "US"),
           !is.na(ARRIVAL_DELAY)) %>%
    group_by(MONTH, DAY, AIRLINE) %>%
    summarise(atrasados = sum(ARRIVAL_DELAY > 10),
              total = n(),
              .groups = "drop")
}
```

QUESTÃO 3:

```r
library(readr)

stats <- read_csv_chunked(
  "flights.csv.zip",
  DataFrameCallback$new(getStats),
  chunk_size = 100000,
  col_types = cols_only(MONTH = col_integer(),
                        DAY = col_integer(),
                        AIRLINE = col_character(),
                        ARRIVAL_DELAY = col_double())
)
```

QUESTÃO 4:

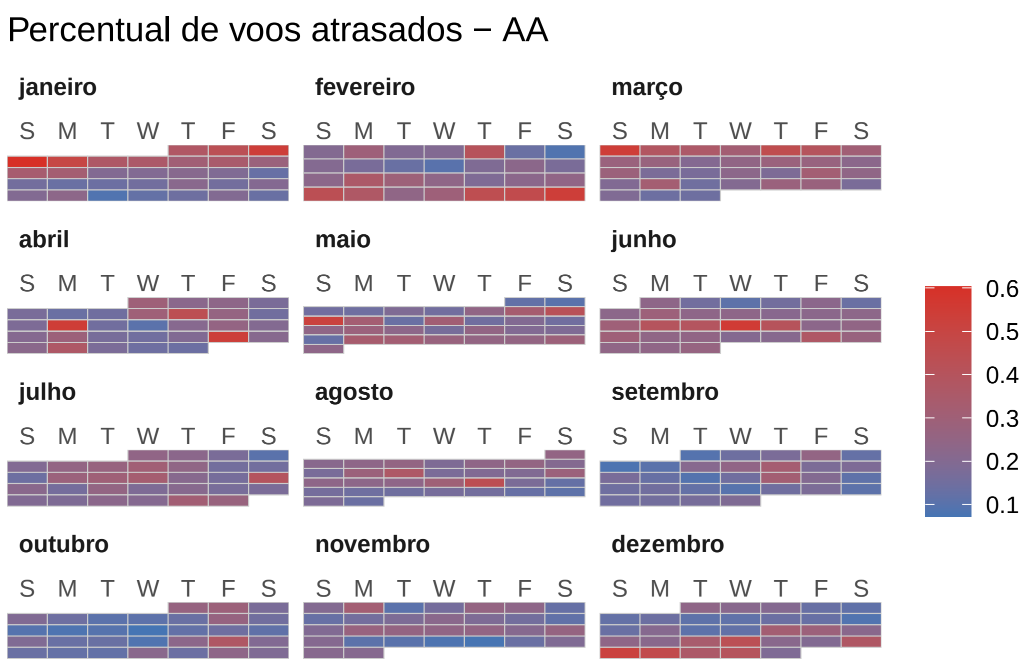
```r
computeStats <- function(stats) {
  stats %>%
    group_by(MONTH, DAY, AIRLINE) %>%
    summarise(atrasados = sum(atrasados),
              total = sum(total),
              .groups = "drop") %>%
    mutate(Cia = AIRLINE,
           Data = as.Date(paste(2015, MONTH, DAY, sep = "-")),
           Perc = atrasados / total) %>%
    select(Cia, Data, Perc)
}

stats <- computeStats(stats)
```

QUESTÃO 5:

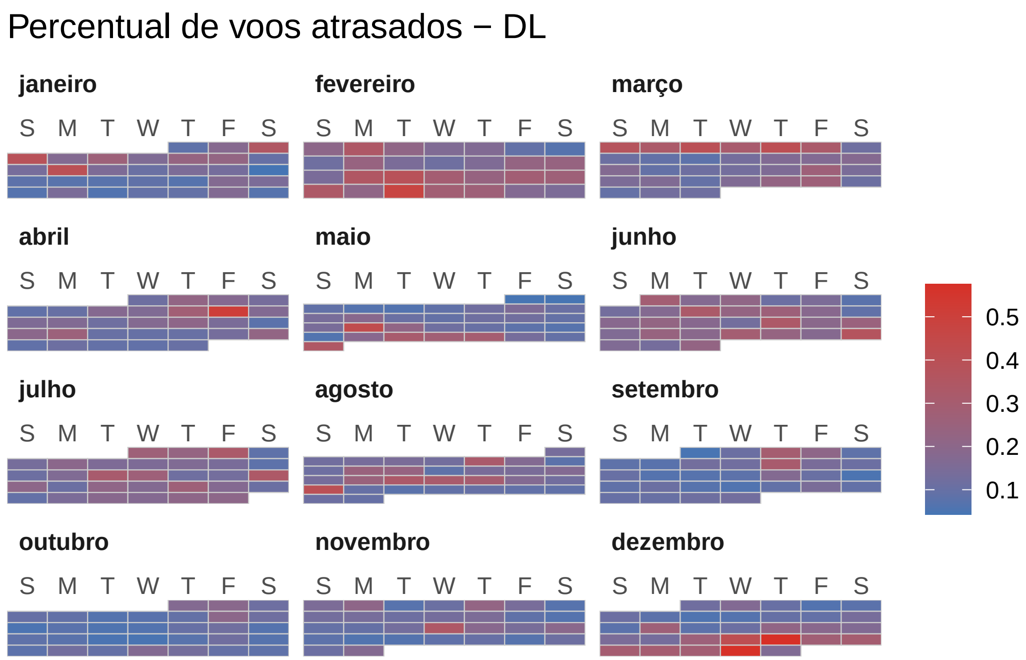
```r
library(devtools)
```

```
Carregando pacotes exigidos: usethis
```

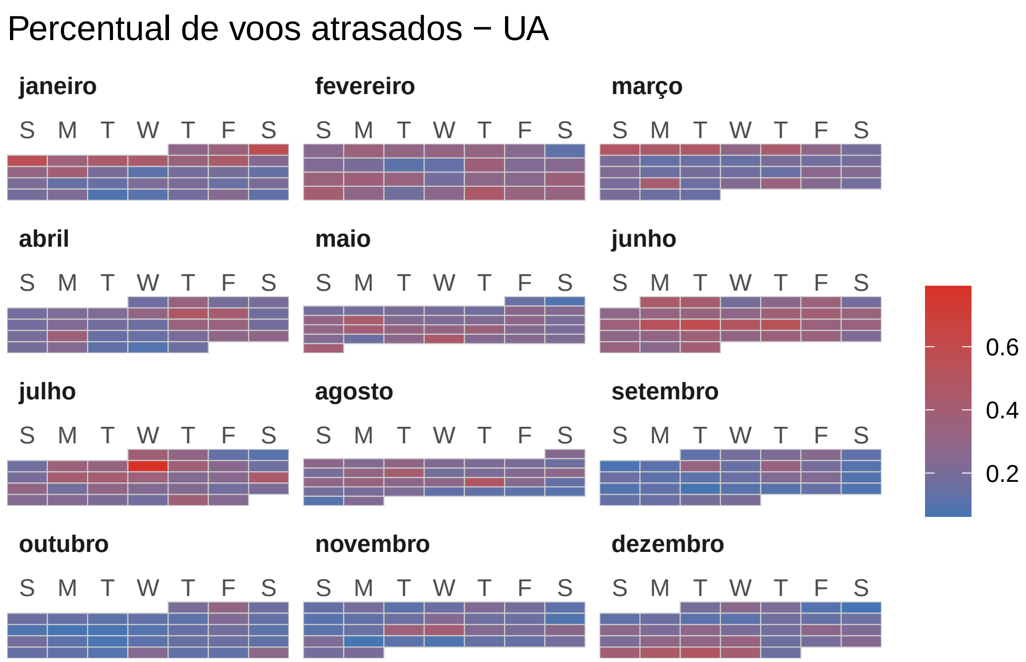
```r
library(ggcal)
library(ggplot2)

pal <- scale_fill_gradient(low = "#4575b4", high = "#d73027")

baseCalendario <- function(stats, cia) {
  dados <- stats %>%
    filter(Cia == cia)

  ggcal(dados$Data, dados$Perc)
}

cAA <- baseCalendario(stats, "AA")
cDL <- baseCalendario(stats, "DL")
cUA <- baseCalendario(stats, "UA")
cUS <- baseCalendario(stats, "US")

cAA +
  pal +
  ggtitle("Percentual de voos atrasados - AA")
```



```
cDL +
  pal +
  ggtitle("Percentual de voos atrasados - DL")
```




```
cUA +
  pal +
  ggtitle("Percentual de voos atrasados - UA")
```




```r
cUS +
  pal +
  ggtitle("Percentual de voos atrasados - US")
```